

Practical No 5

Code 1 :Using Array

Input:

```
#include <iostream>

#include <string>

using namespace std;

#define MAX 100

class ArrayStack {
private:
    string pages[MAX];
    int top;

public:
    ArrayStack() {
        top = -1;
    }

    bool isEmpty() {
        return top == -1;
    }

    bool isFull() {
        return top == MAX - 1;
    }

    void push(string page) {
```

```
    if (isFull()) {  
        cout << "Stack Overflow!\n";  
        return;  
    }  
  
    pages[++top] = page;  
}  
  
string pop() {  
    if (isEmpty()) {  
        return "Stack Underflow";  
    }  
  
    return pages[top--];  
}  
  
string peek() {  
    if (isEmpty()) {  
        return "";  
    }  
  
    return pages[top];  
}  
};  
  
int main() {  
    ArrayStack backStack;  
  
    string currentPage = "Home";
```

```
// Visit pages

cout << "Visiting: " << currentPage << endl;

backStack.push(currentPage);
currentPage = "Google";
cout << "Visiting: " << currentPage << endl;

backStack.push(currentPage);
currentPage = "YouTube";
cout << "Visiting: " << currentPage << endl;

backStack.push(currentPage);
currentPage = "Wikipedia";
cout << "Visiting: " << currentPage << endl;

// Back button

cout << "\nBack button pressed.\n";

currentPage = backStack.pop();

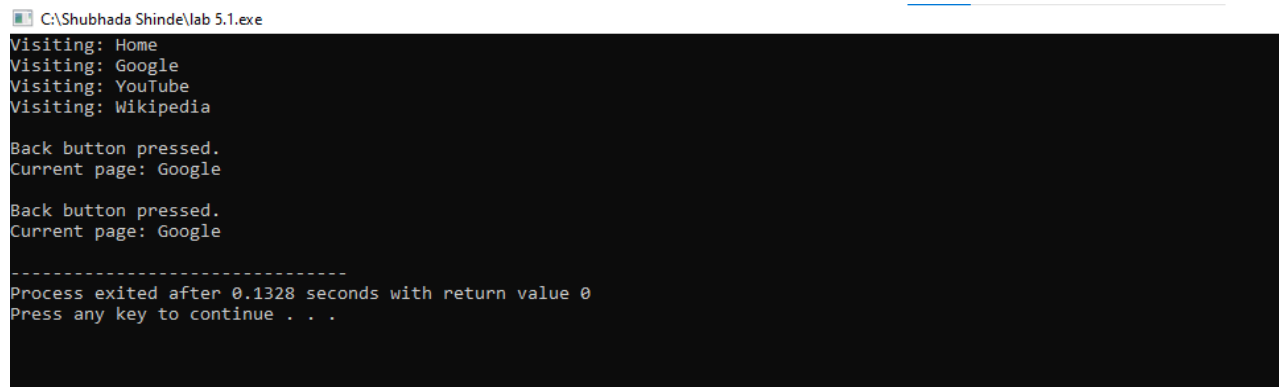
if (!currentPage.empty()) {
    currentPage = backStack.peek();
    cout << "Current page: " << currentPage << endl;
}

cout << "\nBack button pressed.\n";

currentPage = backStack.pop();
```

```
if (!currentPage.empty()) {  
    cout << "Current page: " << currentPage << endl;  
}  
  
return 0;  
}
```

Output:



```
C:\Shubhada Shinde\lab 5.1.exe  
Visiting: Home  
Visiting: Google  
Visiting: YouTube  
Visiting: Wikipedia  
  
Back button pressed.  
Current page: Google  
  
Back button pressed.  
Current page: Google  
  
-----  
Process exited after 0.1328 seconds with return value 0  
Press any key to continue . . .
```

Code 2 : Using Linked List

Input:

```
#include <iostream>

#include <string>

using namespace std;

struct Node {

    string page;

    Node* next;

    Node(string p) {

        page = p;

        next = NULL;

    }

};

class LinkedStack {

private:

    Node* top;

public:

    LinkedStack() {

        top = NULL;

    }

    bool isEmpty() {

        return top == NULL;
```

```
}
```

```
void push(string page) {  
    Node* newNode = new Node(page);  
  
    newNode->next = top;  
    top = newNode;  
}
```

```
string pop() {  
    if (isEmpty()) {  
        return "";  
    }  
  
    Node* temp = top;  
    string page = top->page;  
  
    top = top->next;  
    delete temp;  
  
    return page;  
}
```

```
string peek() {  
    if (isEmpty()) {  
        return "";  
    }  
  
    return top->page;
```

```
}
```

```
~LinkedList() {  
    while (!isEmpty()) {  
        pop();  
    }  
}
```

```
};
```

```
int main() {  
    LinkedList backStack;  
  
    string currentPage = "Home";  
  
    // Visit pages  
    cout << "Visiting: " << currentPage << endl;  
  
    backStack.push(currentPage);  
    currentPage = "Google";  
    cout << "Visiting: " << currentPage << endl;  
  
    backStack.push(currentPage);  
    currentPage = "YouTube";  
    cout << "Visiting: " << currentPage << endl;  
  
    backStack.push(currentPage);  
    currentPage = "Wikipedia";  
    cout << "Visiting: " << currentPage << endl;
```

```
// Back button

cout << "\nBack button pressed.\n";

currentPage = backStack.pop();

cout << "Going back to: " << currentPage << endl;

cout << "\nBack button pressed.\n";

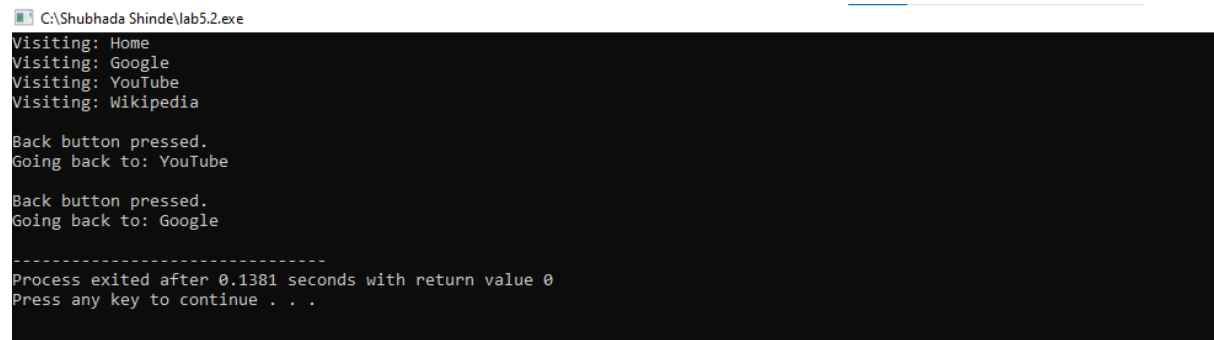
currentPage = backStack.pop();

cout << "Going back to: " << currentPage << endl;

return 0;

}
```

Output:



```
C:\Shubhada Shinde\lab5.2.exe
Visiting: Home
Visiting: Google
Visiting: YouTube
Visiting: Wikipedia

Back button pressed.
Going back to: YouTube

Back button pressed.
Going back to: Google

-----
Process exited after 0.1381 seconds with return value 0
Press any key to continue . . .
```